# ADTA 5900/5750.501: Applied Natural Language Processing

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# Assignment 2

## 1. Overview

The rise of cloud computing has facilitated the emergence of big data. Cloud computing commodifies computing time and data storage using standardized technologies.

Big data is a term for large volumes of data that can be both structured and unstructured. These enormous volumes of data overwhelm the digital world every second. However, it is not the amount of data that is important. What we can do with the data matters: Big data analytics can provide insights that lead to better decisions and strategic moves.

The emergence of cloud computing made it easier to provide the best of technology in the most cost-effective packages. Cloud computing has reduced costs and made a wide array of applications available to companies of all sizes: small, mid-sized, big, and giant corporations.

## 2. Google Cloud Platform (GCP): Service Account

In classwork, various enterprise services of Google Cloud Platform (GCP) will be used. Cybersecurity is one of the top-priority concerns when using cloud services to run applications. To create and run Natural Language Processing (NLP) applications in GCP, the user is required to set up a service account that can be used in GCP's sophisticated authentication system.

**IMPORTANT NOTES**:
*--) All the documents posted on the Canvas page **GOOGLE CLOUD PLATFORM: GCP for Deep Learning – TF2** should be used for HW 1.*

*--) All the documents posted on the Canvas page **GOOGLE CLOUD PLATFORM: GCP for Natural Language Processing (NLP)** should be used for HW 1.*

## 3. PART I: Request GCP Free Credit Coupon and Redeem It (10 Points)

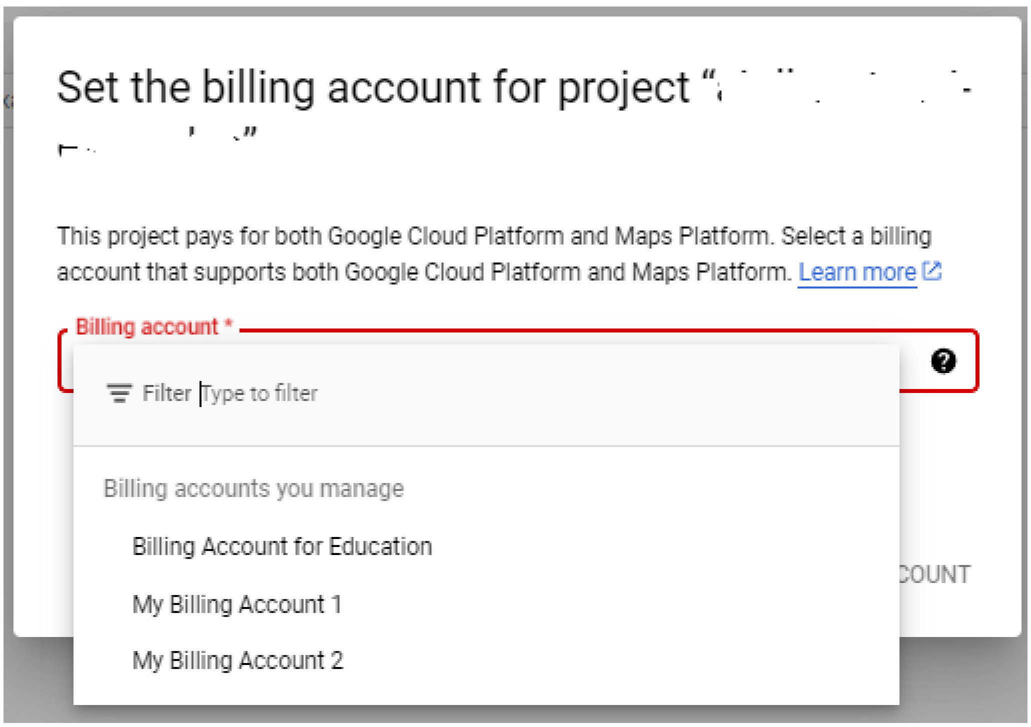
As discussed, from the instructor’s request, Google Cloud Platform (GCP) offers a $50.00 free credit coupon to each student who can redeem and use the credit for classwork.

**TO-DO**

- Access the link provided in the instructor’s announcement email to request a free credit coupon.
- Redeem the free credit coupon following the instructions in GCP’s email.
- Capture the screenshot that shows “Billing Account for Education.”

**SUBMISSION REQUIREMENT PART I #1**:

- Submit the screenshot that shows “Billing Account for Education” like this:




**IMPORTANT NOTES**:

*--) As discussed, by redeeming the coupon, not only does the student use the credit to pay for GCP services to do classwork, but he/she also* ***helps that GCP will continue providing free credit coupons*** *to students in the following semesters.*

## 4. PART II: GCP: Dialogflow CX: Fundamental Concepts (20 Points)

As discussed in the lecture, in GCP: Dialogflow CX, the following concepts are used to design and manage conversation flows:

1. Conversation Session, a.k.a. Conversation or Session
2. Agent
3. Flow
4. Page
5. State
6. State Handler, a.k.a. Handler
7. Intent
8. Entity

**TO-DO**

- Access and read the documentation of GCP: Dialogflow CX, focusing on the above concepts.
    - At this link: https://cloud.google.com/dialogflow/cx/docs

**IMPORTANT NOTES:**
*--) While reading the documentation of GCP: Dialogflow CX, the student should **ignore** the information related to **advanced features** such as **Webhook**.*

**SUBMISSION REQUIREMENT PART II**:
--) Write a report on what the student can learn about each concept mentioned above focusing on the following points:

- What is the concept?
- What role does the concept play in an AI Dialogue System created using Dialogflow CX?

## 5. PART III: GCP: Dialogflow CX: Create a Virtual Agent (30 Points)

Any AI conversational system created using Dialogflow CX requires at least a virtual agent (Ref: GCP: Dialogflow CX documentation).

For PART II, the following references can be used:

- File: gcp_dialogflow_cx_howto_enable_API_create_agent.pdf
- Documentations: GCP Dialogflow CX

**TO-DO**

- Access and open GCP Dialogflow CX console
- Enable the Dialogflow CX API and capture the screenshot showing the API has been enabled
- Create a virtual agent and capture the screenshot showing the agent has been created

**SUBMISSION REQUIREMENT PART III**:
--) Write a report on the steps of completing the tasks with screenshots as illustrations.

## 6. PART IV: GCP: Dialogflow CX: A Simple Conversation (40 Points)

It is assumed that a retail store named Apparel-X sells only one type of clothing: medium-size green shirts for $30.00. There is no tax on any purchase.

Customers are also assumed to know that they can buy only one type of clothing at this store. It means they do not ask about anything else.

Customer C calls the store and has the following conversation with Staff A:

A: Good morning. How can I help you?
C: Good morning. I want to buy a shirt. What color do you have?
A: It's green.
C: Great! How much is it?
A: $30.00
C: What size?
A: Medium. Is it OK for you?
C: Yes. I like it. I'll get to the store and buy one.
A: We're open until 7:00 PM tonight.
C: Thanks. Bye.
A: Thank you very much. Bye.

**Question 3.1:**

- (a) How many turns are there in the above conversation?
- (b) Which turns should be included in "Default Start Flow" (See GCP: Dialogflow CX documentation)?

**Question 3.2**

- Following the information provided in the documentation of GCP: Dialogflow CX, design the conversation flow for the above dialogue.
- Capture a screenshot(s) for each step of the designing process to show what has been done.
- Write a report to discuss each step of designing the conversation flow in detail.

  **IMPORTANT NOTES:**
  *--) It is assumed that customers should have the same conversation when calling the store.*
  *--) The student needs to design ONLY ONE FLOW besides Default Start Flow.*

**Question 3.3**

Based on the conversation design, provide the following pieces of information:

- (a): What conversation session does the design visualize?
- (b): How many pages? What does each page refer to?
- (c): How many states? What does each state refer to?
- (d): Are there any state handlers? If YES, what are they?
- (e): Are there intents? What are they?
- (f): Are there entities? What are they?

**SUBMISSION REQUIREMENT PART IV #1**:
--) Submit the answers to Question 3.1

**SUBMISSION REQUIREMENT PART IV #2**:
--) Provide the design of the conversation flow as required in Question 3.2
--) Submit screenshots of the whole conversation flow design process.
--) Submit the report

**SUBMISSION REQUIREMENT PART IV #3**:
--) Submit the answers to Question 3.3

## 7. HOWTO Submit

The student is required to submit all the sections, i.e., submission requirements, in a Microsoft Word document that is sent to the instructor (Thuan.Nguyen@unt.edu ) as an attachment to a UNT email.

The subject of the email must be:

- "ADTA 5750: Assignment 2 – Submission."

**Due date & time: 11:00 PM – Wednesday 09/25/2024**